

Relative Covariance

Nuno Carvalho

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***Note:** Parts of this document may be AI-written. My general workflow is to write sections of the article myself, laying out the general points I want to hit on and going into detail where relevant. I then converse with an LLM to identify errors in the logic or math and opportunities to use clearer or more generalizable notation. I then have the LLM make the edits to reflect our proposed changes, including adding more detailed derivations.*

This article describes how the phenotypic covariance between two individuals i and j can be modeled by limited information we have about the pair. I derive formulas from the starting trait generative model and only introduce assumptions at the point the derivation requires it. I start with a simple trait architecture and iteratively add more complexity.

1 Additive genetics under random mating with independent environment

We begin with the simplest architecture: a trait determined by an additive genetic component and an independent environmental component, in a randomly mating population.

1.1 The phenotype model

For an individual i we write the (standardized) phenotype as

$$y_i = g_i + e_i, \tag{1}$$

where g_i is the genetic value and e_i the environmental deviation. We take y to be standardized across the population, $E[y] = 0$ and $\text{Var}[y] = 1$, and both components to be mean-centered, $E[g] = E[e] = 0$, with variances $\text{Var}[g] = V_A$ and $\text{Var}[e] = V_E$.

We then introduce the following assumptions.

[Additivity] Additive genetic value

The genetic value is a fixed additive function of the individual's own genotype,

$$g_i = \sum_{k=1}^M \beta_k x_{ik},$$

where β_k is the per-standardized-allele effect of variant k , x_{ik} is the standardized genotype ($E[x_{ik}] = 0$, $\text{Var}[x_{ik}] = 1$), and M is the number of variants in the genome. Non-causal variants are included with $\beta_k = 0$.

Commonly bundled into: the additive model; no dominance, no gene–gene interactions (GxG, epistasis), and no gene–environment interactions (GxE) — each would make the genetic value depend on more than the individual’s own genotype in an additive way.

[GE-indep] Gene–environment independence

An individual’s environmental component is independent of their own genetic component:

$$g_i \perp e_i$$

Commonly bundled into: no gene–environment correlation or interaction, within an individual.

[No-IGE] No indirect genetic effects

One individual’s genetic component is independent of another individual’s environmental component:

$$g_i \perp e_j \quad (i \neq j).$$

Commonly bundled into: no indirect (e.g. parental) genetic effects — one individual’s genotype does not shape another’s environment.

[Env-indep] Independent environments

The environmental component is independent across individuals, including relatives:

$$e_i \perp e_j \quad (i \neq j).$$

Commonly bundled into: no shared/common environment (C); no environmental confounder shared by the pair (e.g. population stratification).

Because $g_i \perp e_i$ ([GE-indep]), $\text{Cov}[g_i, e_i] = 0$ and the phenotypic variance decomposes as $\text{Var}[y] = V_A + V_E$. Narrow-sense heritability is typically defined as:

$$h^2 = \frac{V_A}{\text{Var}[y]}$$

Since $\text{Var}[y] = 1$, $h^2 = V_A$ under this model.

1.2 Phenotypic covariance is genotypic covariance

For a single fixed pair, y_i and y_j are two fixed values, so “their covariance” only has meaning as an average over an ensemble of pairs. The ensemble we use is the set of all pairs that share a given piece of *relatedness information* $\mathcal{I}_{ij}$ about the pair, and the object of interest is the conditional cross-product

$$C_{ij}(\mathcal{I}) := \mathbb{E}[y_i y_j \mid \mathcal{I}_{ij}]. \quad (2)$$

If $\mathbb{E}[y_i \mid \mathcal{I}] = 0$, then $C_{ij}(\mathcal{I}) = \text{Cov}[y_i, y_j \mid \mathcal{I}]$. For most information $\mathcal{I}$, like the realized relatedness or degree between two relatives, this equivalence is applicable. An exception is when the information we are modelling contains individuals’ actual genotype information, for which $\mathbb{E}[y_i \mid \text{genotypes}] = g_i \neq 0$, therefore the expected cross-product of two individuals is not equivalent to their covariance given the fixed information.

Using (1) together with [No-IGE] and [Env-indep], every cross term involving e vanishes in expectation (each factors through a mean-zero environmental term), so the phenotypic cross-product reduces to the genetic one:

$$C_{ij}(\mathcal{I}) = \mathbb{E}[(g_i + e_i)(g_j + e_j) \mid \mathcal{I}] = \mathbb{E}[g_i g_j \mid \mathcal{I}]. \quad (3)$$

The rest of the article evaluates (3) at four levels of resolution for $\mathcal{I}$.

1.3 Level 0 — the exact genetic cross-product

At the finest resolution we condition on the realized genotypes of i and j at the causal variants, for which we know the true causal effect sizes. Then g_i and g_j are fixed, and by [Additivity],

$$C_{ij} = \mathbb{E}[y_i y_j \mid \mathbf{x}_i, \mathbf{x}_j, \boldsymbol{\beta}] = g_i g_j = \sum_k \sum_{k'} \beta_k \beta_{k'} x_{ik} x_{jk'} \quad (4)$$

Under our model and assumptions so far, this returns the true expected cross-product of two individuals’ phenotypes. The genetic cross-product between two individuals is, fundamentally, an effect-size-weighted function of the genotypes they actually carry at causal sites. Notably, this formula does not require any knowledge of the relatedness of the two individuals.

1.4 Level 1 — genome-wide allelic sharing

If we do not know the causal effect sizes, then we cannot apply (4). However, we can assume that genetic effects’ cross-products are independent of individuals’ allelic sharing patterns.

[Random-effects] Random effects, exchangeable across variants

Treating the effects as random, they are mean-zero and uncorrelated across variants, the distribution of their cross-products $\{\beta_k\beta_{k'}\}$ is independent of the genotypes $\mathbf{X}$, and every variant carries the same second moment:

$$\mathbb{E}[\beta_k] = 0, \quad \mathbb{E}[\beta_k\beta_{k'}] = 0 \quad (k \neq k'), \quad \mathbb{E}[\beta_k\beta_{k'} \mid \mathbf{X}] = \mathbb{E}[\beta_k\beta_{k'}], \quad \mathbb{E}[\beta_k^2] = \mathbb{E}[\beta_{k'}^2].$$

The last condition is a statement about the *prior* we place on effects, not about their realized values: it says we have no information distinguishing one variant from another, so we weight them symmetrically.

Commonly bundled into: the infinitesimal / random-effects model.

Taking the expectation of (4) over the effects, and using the independence of the cross-products from the genotypes ([Random-effects]) to replace $\mathbb{E}[\beta_k\beta_{k'} \mid \mathbf{X}]$ by $\mathbb{E}[\beta_k\beta_{k'}]$:

$$\mathbb{E}[g_i g_j \mid \mathbf{x}_i, \mathbf{x}_j] = \sum_k \sum_{k'} \mathbb{E}[\beta_k\beta_{k'}] x_{ik} x_{jk'}.$$

Because the effects are mean-zero and uncorrelated across variants, $\mathbb{E}[\beta_k\beta_{k'}] = 0$ whenever $k \neq k'$; the entire off-diagonal ($k \neq k'$) collapses regardless of any correlation between the genotypes. What remains is the diagonal, weighted by each variant's effect variance:

$$\mathbb{E}[g_i g_j \mid \mathbf{x}_i, \mathbf{x}_j] = \sum_k \mathbb{E}[\beta_k^2] x_{ik} x_{jk}.$$

The additive variance is the total of these: since $\text{Var}[x_{ik}] = 1$ and the off-diagonal effect terms vanish, $V_A = \text{Var}[g_i] = \sum_k \mathbb{E}[\beta_k^2]$, and thus $\mathbb{E}[\beta_k^2] = V_A/M$. Effectively, the per-variant effect weighting of (4) has been replaced by a uniform weighting across all variants. Therefore

$$C_{ij}(\mathbf{x}_i, \mathbf{x}_j) = V_A \cdot \frac{1}{M} \sum_k x_{ik} x_{jk}. \quad (5)$$

The per-variant effect weighting of (4) has been replaced by the genome-wide genotype cross-product (i.e. allelic sharing), which we'll define by:

$$A_{ij} = \frac{1}{M} \sum_k x_{ik} x_{jk}$$

This can be thought of as an identity-by-state (IBS)-based measure of “relatedness”, and is exactly the (i, j) entry of what is typically computed as a Genetic Relatedness Matrix (GRM), $\mathbf{A}$. Therefore, we can further simplify:

$$\boxed{C_{ij}(A_{ij}) = V_A A_{ij}} \quad (6)$$

1.5 Level 2 — realized relatedness

The problem with calculating A_{ij} is that we may not have genotyped every causal variant that contributes to a trait. As long as some of the causal variants are not in perfect

correlation with a genotyped variant, then the covariance of two individuals' phenotypes will not fully take into account the contributions of the ungenotyped causal variants. However, when individuals are related, their IBS-based relatedness from genotyped variants is informative of the IBS-based relatedness of ungenotyped variants. Therefore, the former can be used to capture the contributions of the latter.

When individuals inherit a genomic segment from a common ancestor, then all of the variants within that genomic segment will be identical-by-state, as long as we assume that de-novo mutations have not caused the inherited alleles to be different. These genomic segments and their variants are said to be identical-by-descent (IBD).

[No-mutation] IBD implies IBS

Alleles that are identical-by-descent are also identical-by-state: descent is transmitted without mutation.

Commonly bundled into: the infinite-sites / no-recurrent-mutation assumption.

When two individuals' alleles are IBD, then they are always IBS; when they are not IBD, they can still be IBS. Therefore, we can decompose the IBS-based relatedness A_{ij} into a sum over IBD allele pairs and a sum over non-IBD allele pairs. Write each standardized genotype as the sum of its two per-allele (maternal and paternal) standardized values, $x_{ik} = z_{ik}^{(m)} + z_{ik}^{(p)}$. Since $\text{Var}[x_{ik}] = 1$ and the two alleles are independent ([No-inbreeding]), each per-allele value has $E[z] = 0$ and $\text{Var}[z] = \frac{1}{2}$. The genotype cross-product then expands over the four ordered allele pairs, and A_{ij} splits by IBD status:

$$\begin{aligned} A_{ij} &= \frac{1}{M} \sum_k x_{ik} x_{jk} = \frac{1}{M} \sum_k \sum_{u,v \in \{m,p\}} z_{ik}^{(u)} z_{jk}^{(v)} \\ &= \underbrace{\frac{1}{M} \sum_{\text{IBD pairs}} z_{ik}^{(u)} z_{jk}^{(v)}}_{\text{IBD}} + \underbrace{\frac{1}{M} \sum_{\text{non-IBD pairs}} z_{ik}^{(u)} z_{jk}^{(v)}}_{\text{non-IBD}}. \end{aligned} \tag{7}$$

For an IBD allele pair, the two per-allele values are copies of the same ancestral allele and, by [No-mutation], of the same state; they are therefore equal, so their expected cross-product is $E[(z_{ik}^{(u)})^2] = \text{Var}[z_{ik}^{(u)}] = \frac{1}{2}$. For non-IBD allele pairs, we have to make an assumption on how they are distributed.

[Non-IBD-indep] Independence of non-IBD alleles

Any two alleles that are *not* IBD are independent draws from the population allele distribution.

Commonly bundled into: between individuals: random mating, no population structure, and no assortative mating.

[No-inbreeding] No inbreeding

Within an individual, the two homologous alleles are not IBD with each other, and hence (by [Non-IBD-indep]) independent. This results in each variant's genotype variance being $2p_k(1 - p_k)$ — so the standardized x_{ik} has variance 1 — and it keeps the number of alleles a pair shares IBD at a variant in $\{0, 1, 2\}$.

Commonly bundled into: a randomly mating, outbred population (Hardy–Weinberg within individuals).

By the definition of independent draws, the expected cross-product of a non-IBD allele pair is therefore $E[z_{ik}^{(u)}]E[z_{jk}^{(v)}] = 0$. Letting $\text{IBD}_{ij}(k) \in \{0, 1, 2\}$ denote the number of alleles the pair shares IBD at variant k , exactly $\text{IBD}_{ij}(k)$ of the four ordered allele pairs are IBD — each contributing $\frac{1}{2}$ — and the rest contribute 0, so

$$E[x_{ik}x_{jk} \mid \text{IBD}_{ij}(k)] = \frac{1}{2}\text{IBD}_{ij}(k).$$

A variant thus contributes expected cross-product 0, $\frac{1}{2}$, or 1 according to whether the pair shares 0, 1, or 2 alleles IBD there. We define the pair's *realized relatedness* π_{ij} as the average IBD state across variants, halved — equivalently, the fraction of the pair's genome that is IBD:

$$\pi_{ij} := \frac{1}{2M} \sum_k \text{IBD}_{ij}(k). \quad (8)$$

Conditioning on the pair's realized IBD states and using the per-variant identity $E[x_{ik}x_{jk} \mid \text{IBD}_{ij}(k)] = \frac{1}{2}\text{IBD}_{ij}(k)$, linearity of expectation gives the expected allelic sharing:

$$E[A_{ij} \mid \{\text{IBD}_{ij}(k)\}] = \frac{1}{M} \sum_k \frac{1}{2} \text{IBD}_{ij}(k) = \frac{1}{2M} \sum_k \text{IBD}_{ij}(k) = \pi_{ij}.$$

This depends on the IBD states only through π_{ij} , so we may condition on π_{ij} alone. Crucially, it required no assumption that the per-variant IBD states are independent — and indeed they are not: an IBD state changes only at a cross-over within a co-inherited segment, so physically linked variants have strongly correlated IBD states. Linearity of expectation is insensitive to this correlation.

Combining with (6), the pair's expected phenotypic cross-product is their realized relatedness scaled by the additive variance:

$$\boxed{E[y_i y_j \mid \pi_{ij}] = V_A E[A_{ij} \mid \pi_{ij}] = V_A \pi_{ij}}. \quad (9)$$

1.6 Level 3 — genetic degree

In cases where we cannot measure π_{ij} based on per-variant IBD states, we can still estimate it if we know the genetic degree d_{ij} between the individuals.

[Genetic-degree] Degree determines expected relatedness

The genetic degree d_{ij} fixes the distribution of IBD sharing through Mendelian segregation, so that the expected relatedness given the degree is a single number $\bar{\pi}_{d_{ij}}$:

$$E[\pi_{ij} \mid d_{ij}] = \bar{\pi}_{d_{ij}}.$$

Furthermore, under the outbreeding assumption ([No-inbreeding]) — applied here not only to i and j but to the entire pedigree connecting them, so that it contains no inbreeding loops — $\bar{\pi}_{d_{ij}} = 2^{-d_{ij}}$, which is not fully derived here. We can then obtain the expected phenotypic covariance given the degree d_{ij} :

$$E[y_i y_j \mid d_{ij}] = E[E[y_i y_j \mid \pi_{ij}] \mid d_{ij}] = V_A E[\pi_{ij} \mid d_{ij}] = V_A \bar{\pi}_{d_{ij}}. \quad (10)$$

1.7 Summary

Each level conditions on coarser information than the one above and is its conditional expectation; the rightmost column lists the assumption(s) newly required to descend one step.

Explicitly, each level is the conditional expectation of the finer level above it, averaging out one additional layer of information:

$$\begin{aligned} E[y_i y_j \mid \mathbf{x}_i, \mathbf{x}_j] &= E[g_i g_j \mid \mathbf{x}_i, \mathbf{x}_j] = V_A A_{ij}, \\ E[y_i y_j \mid \pi_{ij}] &= E[V_A A_{ij} \mid \pi_{ij}] = V_A \pi_{ij}, \\ E[y_i y_j \mid d_{ij}] &= E[V_A \pi_{ij} \mid d_{ij}] = V_A \bar{\pi}_{d_{ij}}, \end{aligned}$$

starting from the Level-0 value $E[y_i y_j \mid \mathbf{x}_i, \mathbf{x}_j, \boldsymbol{\beta}] = g_i g_j$. Equivalently, the ladder telescopes into a single nested expectation,

$$E[y_i y_j \mid d_{ij}] = E\left[E\left[E[g_i g_j \mid \mathbf{x}_i, \mathbf{x}_j] \mid \pi_{ij}\right] \mid d_{ij}\right] = V_A \bar{\pi}_{d_{ij}}.$$

This is valid down the generative chain $d_{ij} \rightarrow \pi_{ij} \rightarrow (\mathbf{x}_i, \mathbf{x}_j) \rightarrow y$: conditioned on the finer variable, the phenotype does not depend further on the coarser one, so the tower property applies at each step.

Level	Conditioned on	$E[y_i y_j \mid \cdot]$	New assumptions
0	causal genotypes	$g_i g_j$	[Additivity] (given [No-IGE], [Env-indep])
1	realized sharing A_{ij}	$V_A A_{ij}$	[Random-effects]
2	realized relatedness π_{ij}	$V_A \pi_{ij}$	[No-mutation], [Non-IBD-indep], [No-inbreeding]
3	genetic degree d_{ij}	$V_A \bar{\pi}_{d_{ij}}$	[Genetic-degree]

Next architectures restart from Level 0 and re-introduce a subset of the assumptions above, relaxing others: e.g. a shared environment breaks [**Env-indep**]; dominance/epistasis breaks [**Additivity**]; assortative mating and population structure break [**Non-IBD-indep**] (correlating non-IBD alleles) and drive a wedge between realized and pedigree relatedness.

References